

Current Resistance and Ohms Law

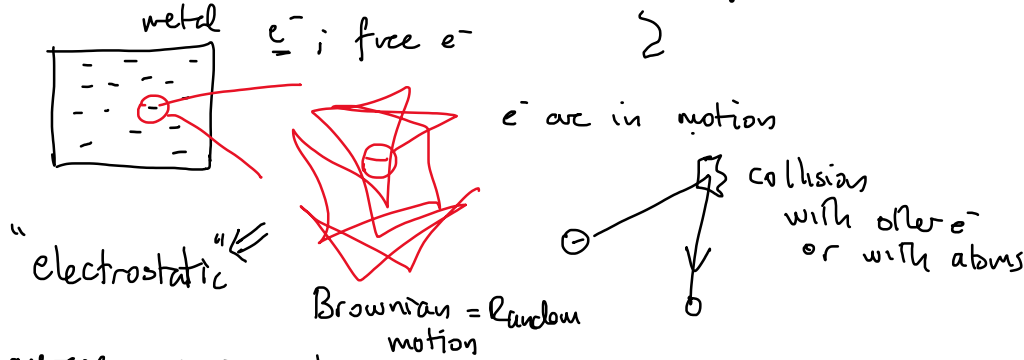
Wednesday, April 3, 2024 10:37 AM

upto now electrostatic $\equiv$ in equilibrium
charges have stopped "moving"

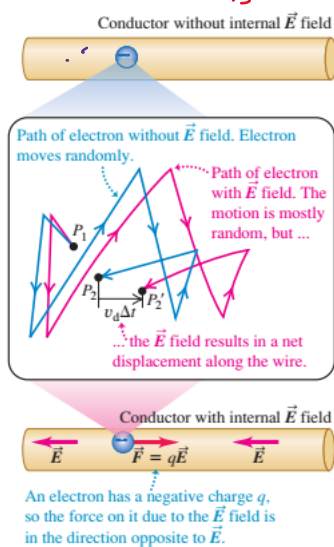
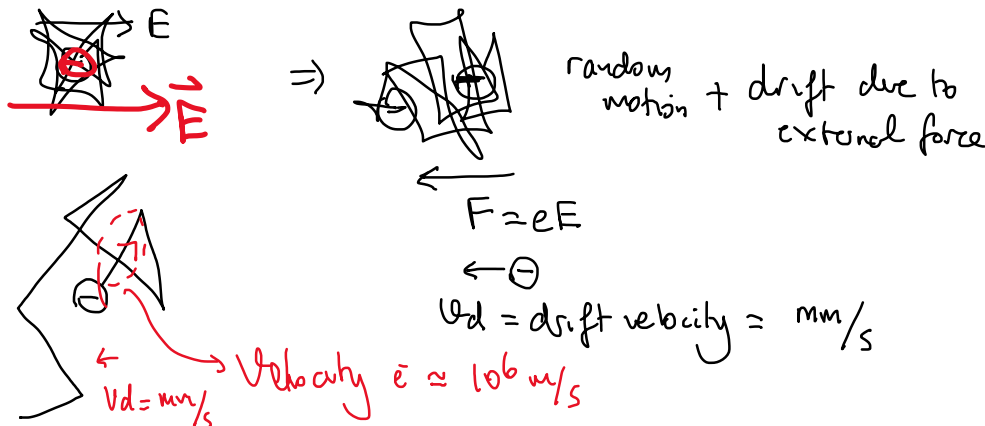


charges are being moved! by external $\vec{E}$ field.

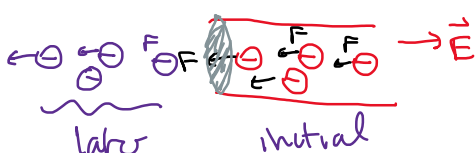
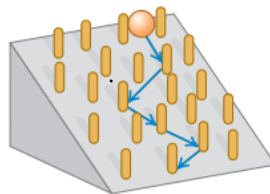
what's the situation in a metal ~~is~~ at equilibrium



on average e^- seem to be motionless; no net drift/motion in any direction
↓
stillklene



25.27 The motion of a ball rolling down an inclined plane and bouncing off pegs in its path is analogous to the motion of an electron in a metallic conductor with an electric field present.



charges moved at certain time.
 $I \equiv \frac{\Delta Q}{\Delta t}$ current

$\frac{4e^-}{1\text{ns}} = I$
 $\Delta t = 1\text{ns}$
 $\text{Ampere} = \frac{\text{Coulomb}}{\text{sec}} = \left[\frac{\text{C}}{\text{s}} = \text{A} \right]$

if ~~the~~ $4e^-$ moves in 1ns $I = ?$

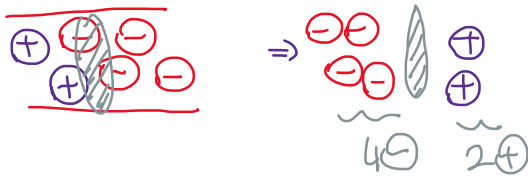
$$I = \frac{Q}{t} = \frac{4 \times 1.6 \times 10^{-19} \text{C}}{1 \times 10^{-9} \text{s}} = 6.4 \times 10^{-10} \text{A}$$

daily life $I = \text{mA}$

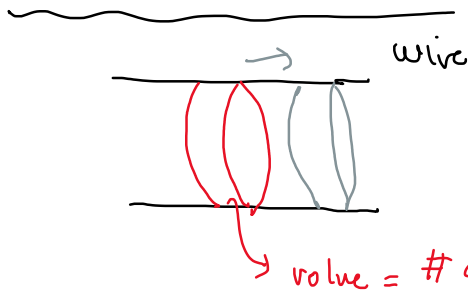
$\leftarrow \ominus \ominus \ominus \equiv \oplus \oplus \oplus \rightarrow$
 $\xrightarrow{E} \oplus \rightarrow F = qE$

reality convention = habit direction of current = direction of $\oplus$ charges

ex



$$I = \frac{4\ominus + 2\oplus}{\Delta t} = \frac{6 \text{ charges}}{\Delta t}$$



$$I = \frac{\Delta Q}{\Delta t}$$

$$q_e = 1.6 \times 10^{-19} \text{C}$$



Area $\sim A$ $\Delta V = \text{volume} = A \Delta x$

no of charges = $(\Delta V) n$

$$I = \frac{\Delta Q}{\Delta t} = \frac{q_e \cdot \text{no of charges}}{\Delta t} = \frac{q_e n A \Delta x}{\Delta t}$$

$\frac{\text{charge}}{\text{volume}} = \text{const.}$

$\frac{\Delta x}{\Delta t} = v_d = \text{drift velocity}$

$$I = q_e n A v_d$$

$$q_e = 1.6 \times 10^{-19} \text{C}$$

$n = \text{const} = \text{no of charge} = \text{charge density}$

$$J \equiv \frac{I}{A} = q_e n v_d$$

current density = flux

$$J = \frac{I}{A} \Rightarrow \left(\begin{array}{c} \xrightarrow{E} \\ \xrightarrow{J} \end{array} \right)$$

$$\uparrow J, I \uparrow \propto \vec{E} \uparrow$$

$$J = \sigma E \quad \text{OHM'S LAW}$$

$$J = \sigma E$$

conductivity
const.

$$\Delta V = -\int \vec{E} \cdot d\vec{r} = -EL = EL$$

$$\frac{I}{A} = J = \sigma E = \sigma \frac{\Delta V}{L}$$

ohm's
Law

σ = conductivity
(sigma)

resistivity
const. $\rho = \frac{1}{\sigma}$
rho

$$\Delta V = I \frac{L}{A \sigma}$$

$$\Delta V = I \frac{L}{A} \rho$$

ohm's Law

Resistance R

$$\Delta V = IR$$

ohm's Law

Table 25.1 Resistivities at Room Temperature (20°C)

Substance	ρ ($\Omega \cdot m$)	Substance	ρ ($\Omega \cdot m$)
Conductors		Semiconductors	
Metals		Pure carbon (graphite)	3.5×10^{-5}
Silver	1.47×10^{-8}	Pure germanium	0.60
Copper	1.72×10^{-8}	Pure silicon	2300
Gold	2.44×10^{-8}	Insulators	
Aluminum	2.75×10^{-8}	Amber	5×10^{14}
Tungsten	5.25×10^{-8}	Glass	$10^{10} - 10^{14}$
Steel	20×10^{-8}	Lucite	$> 10^{13}$
Lead	22×10^{-8}	Mica	$10^{11} - 10^{15}$
Mercury	95×10^{-8}	Quartz (fused)	75×10^{16}
Alloys		Sulfur	10^{15}
Manganin (Cu 84%, Mn 12%, Ni 4%)	44×10^{-8}	Teflon	$> 10^{13}$
Constantan (Cu 60%, Ni 40%)	49×10^{-8}	Wood	$10^8 - 10^{11}$
Nichrome	100×10^{-8}		

25.9 This resistor has a resistance of 5.7 k Ω with a precision (tolerance) of $\pm 10\%$.

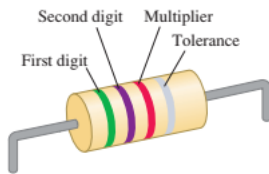


Table 25.3 Color Codes for Resistors

Color	Value as Digit	Value as Multiplier
Black	0	1
Brown	1	10
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9

$$I = \frac{Q}{t}$$

$$J = \frac{I}{A} = \sigma E = \frac{E}{\rho}$$

$$I = q_e n A v_d$$

$$q_e = 1.6 \times 10^{-19} C$$

n = charge density

A = crosssectional area

v_d = drift velocity

J = current density

σ = conductivity

ρ = resistivity

$$\Delta V = IR ; R = \rho \frac{L}{A}$$

volt = Ampere ohm

resistance

$$[V] = [A][\Omega]$$

$$R = [\Omega] = \frac{\text{volt}}{\text{Ampere}}$$

$$R = \rho \frac{L}{A} \Rightarrow [\Omega] = \rho \left[\frac{m}{m^2} \right]$$

$$\rho = [\Omega m]$$

ex) Copper wire diameter of 1.02 mm

free e^- charge density $n = 8.5 \times 10^{28} \frac{1}{m^3}$

$$I = 1.67 A$$

a) $J = ?$

$$J = \frac{I}{A} = \frac{1.67 A}{3.14 (0.51)^2 \times 10^{-6} m^2} = 2.6 \times 10^6 A/m^2$$

$$\text{area} = \pi \left(\frac{d}{2} \right)^2$$

b) $v_d = ?$

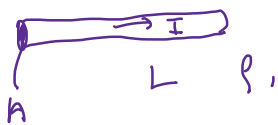
$$I = q n A v_d$$

$$u_d = \frac{1.67}{(1.6 \times 10^{-19}) (8.5 \times 10^{28}) 3.14 (0.51)^2 \times 10^{-6}} \left[\frac{A}{C \frac{1}{m^3} m^2} \right]$$

$$u_d = 0.15 \text{ mm/s} = 1.5 \times 10^{-4} \text{ m/s}$$

$$\frac{C/s}{C \frac{1}{m}} = \left[\frac{m}{s} \right]$$

ex) Area wire = $8.2 \times 10^{-7} \text{ m}^2$ $L = 50 \text{ m}$
Copper $\rho = 1.72 \times 10^{-8} \Omega \text{ m}$ $I = 1.67 \text{ A}$



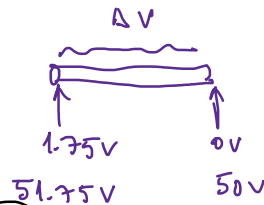
a) $R = ?$

$$R = \rho \frac{L}{A} = \frac{1.72 \times 10^{-8} \times 50}{8.2 \times 10^{-7}} \Omega \text{ m} \frac{m}{m^2}$$

$$R = 1.05 \Omega$$

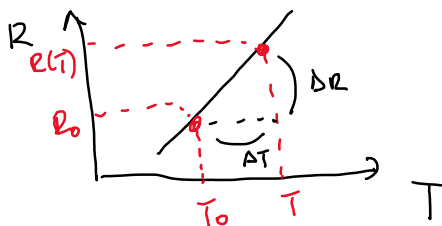
b) $E = ?$ $E = \frac{V}{L} = \rho J = 1.72 \times 10^{-8} \frac{1.67 \text{ A}}{8.2 \times 10^{-7} \text{ m}^2} = 0.035 \frac{\text{N}}{\text{C}}$
 $E = \frac{\Delta V}{L} = 0.035 \frac{\text{V}}{\text{m}}$

c) $\Delta V = EL = IR = 1.75 \text{ V}$



Resistance changes with temperature ?! w ?

$$R(T) = R_0 (1 + \alpha (T - T_0)) \Rightarrow \frac{\Delta R}{\Delta T} = \text{slope}$$



ex) $\alpha = 0.0039 \frac{1}{^\circ\text{C}} ; \frac{1}{\text{K}}$

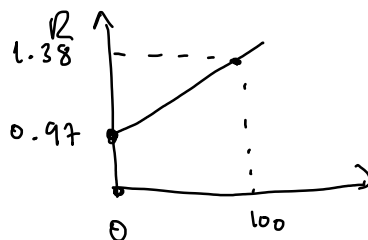
if $R @ 25^\circ\text{C} = 1.05 \Omega$

$R @ 0^\circ\text{C} = ?$

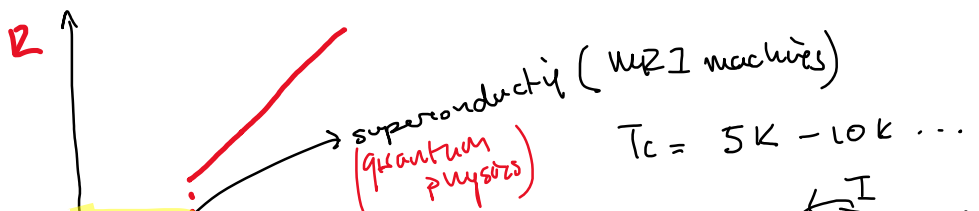
$100^\circ\text{C} = ?$

$$R(100) = 1.05 (1 + 0.0039 (100 - 25)) = 1.38 \Omega$$

$$R(0) = 1.05 (1 + 0.0039 (0 - 25)) = 0.97 \Omega$$



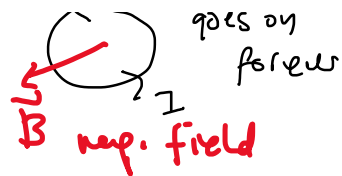
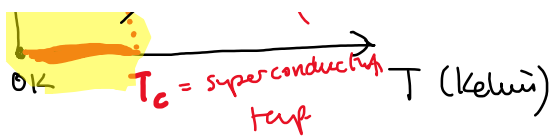
if we cool it way down?



superconductivity (MRI machines)
(quantum physics)

$$T_c = 5 \text{ K} - 10 \text{ K} \dots$$

$$\leftarrow I$$



$T_c \sim 180 K$ (some materials)
 $\hookrightarrow$ quite expensive

Electro Motor Force (EMF)

EMF makes current flow $\equiv$ Voltage difference

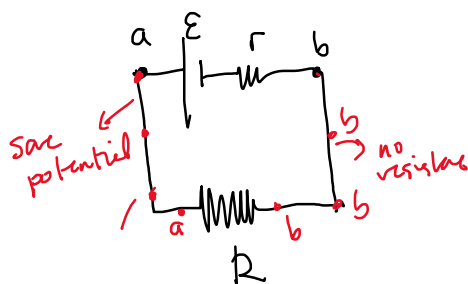
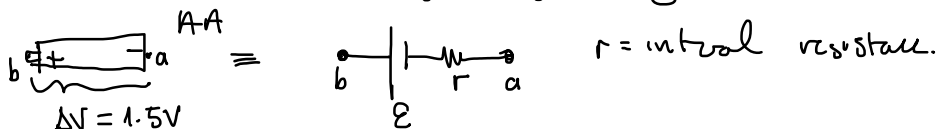
Batteries are sources of EMF.



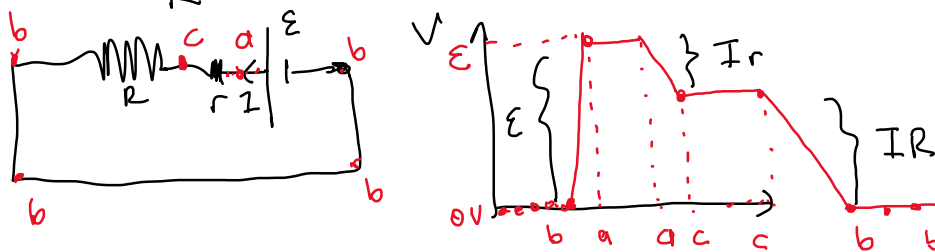
EMF unit is Volts!

Ideal battery has no internal resistance

but in reality every battery has internal resistance



$\hookrightarrow$ wires don't have resistance = ideal



$$\sum \text{potential rise } b \rightarrow a = \sum \text{potential decrease}$$

$$E = Ir + IR \Rightarrow I = \frac{E}{r + R}$$

$$\begin{aligned} E &= 12V \\ r &= 2\Omega \\ R &= 4\Omega \end{aligned}$$

$$I = \frac{12}{2+4} = 2A$$

midterm

POWER

$$\text{Power} = \frac{\text{Energy}}{\text{time}} \Rightarrow [\text{Watt}] = \left[\frac{J}{s} \right]$$

$$P = \frac{\text{Energy}}{t} = \frac{\Delta U}{t} = \frac{q \Delta V}{t} \quad \text{①} \quad \Delta V = \frac{q}{I} \Delta V$$

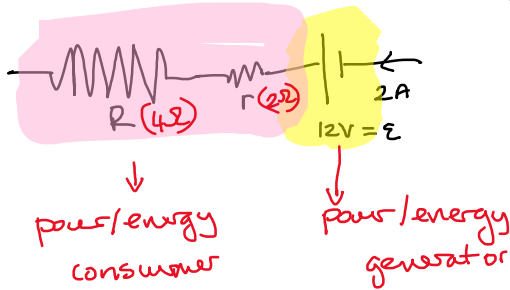
$$P = I \Delta V \quad ; \quad \Delta V = IR$$

$$P = I^2 R = \frac{(\Delta V)^2}{R} = I \Delta V \quad \Rightarrow \text{watt} = \frac{J}{s}$$

$$W \cdot s = J$$

Electric bill $\Rightarrow$ kw.hr
energy

$$\text{Elec. bill} \Rightarrow \text{kw.hr} = 1000 \times 3600 \frac{Ws}{J} = 3.6 \times 10^6 J = 1 \text{ kw.hr}$$



$$I^2 R = -16W \quad I^2 r = -8W \quad I \Delta V = +24W \quad = 0$$

c) if we run an oven @ 240V $I = 20A$ for 4hrs.

$$P = ? \quad P = I \Delta V = 240(20) = 4.8 \text{ kW}$$

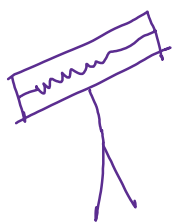
$$\text{Energy} \quad P \Delta t = U = 4.8 \text{ kW} \cdot 4 \text{ hrs}$$

$$U = 19.2 \text{ kw.hr}$$

$$\text{how much} \quad 1 \text{ kw.hr} = 2.2 \text{ TL}$$

$$U \approx 40 \text{ TL}$$

cook everyday $\Rightarrow 40 \times 30 = 1200 \text{ TL}$. monthly cost of oven.



WFO (thermal heating)

$$P = 2 \text{ kW}$$

for heating each day 6hrs
month 180 hrs

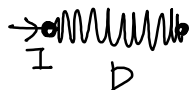
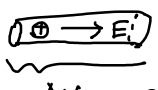
$$U = 2 \text{ kW} \cdot 180 \text{ hr} = 3600 \text{ kw.hr}$$

$$2.2 \text{ TL}$$

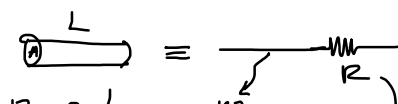
expensive!! $\approx 8000 \text{ TL}$

End of Ch. Ohm's Law.

DC circuits



reality
copper wire

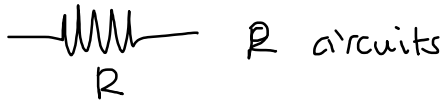
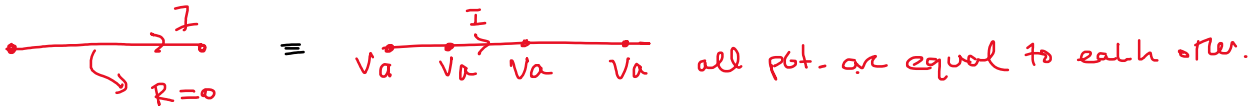


$$\Delta V = IR$$

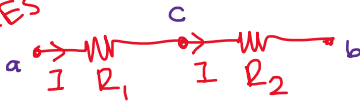
$$R = \rho \frac{L}{A} \quad \text{resistance}$$

in our phys. lectures

wires $R \approx 0$



SERIES



$$V_{ac} + V_{cb} = V_{ab}$$

$$IR_1 + IR_2 = IR_{eq}$$

$$IR_1 + IR_2 = IR_{eq}$$



$$\Rightarrow V_{ab} = IR_{eq}$$

$$R_{eq} = R_1 + R_2$$

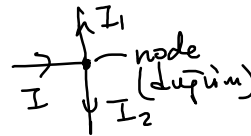


$$R_{eq} = R_1 + R_2 + \dots + R_N$$

PARALLEL



$$V_{ab} = IR_1 = IR_2$$

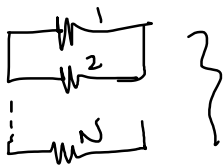


$$V_{ab} = IR_{eq}$$

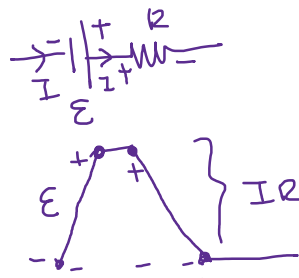
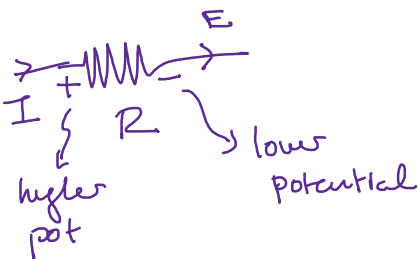
$$I = I_1 + I_2$$

$$\frac{V_{ab}}{R_{eq}} = \frac{V_{ab}}{R_1} + \frac{V_{ab}}{R_2}$$

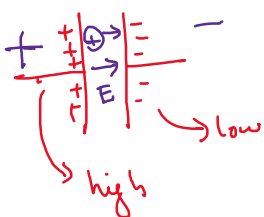
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



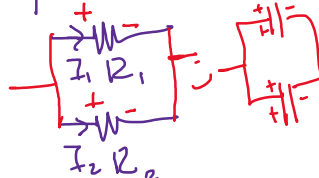
$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N} \right)^{-1}$$



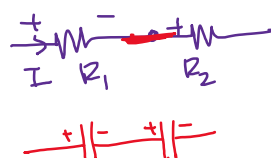
current enters to the higher potential part of a resistance.

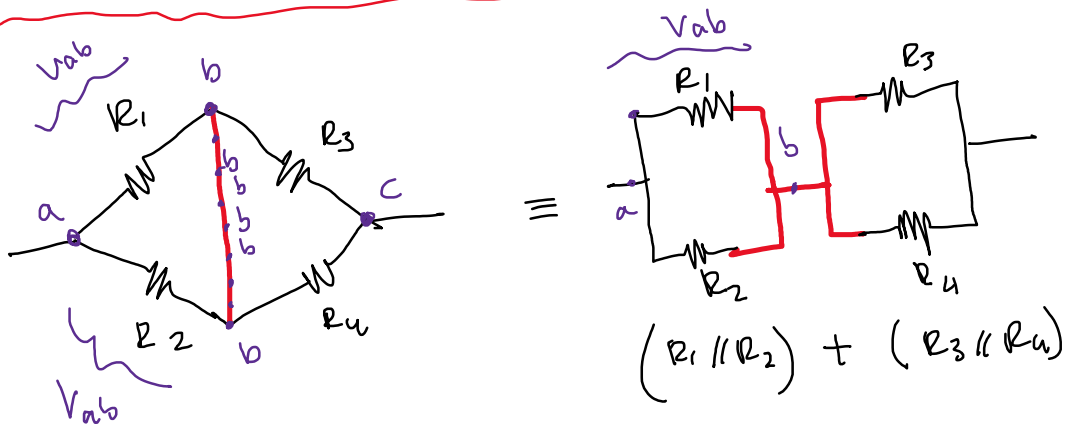


parallel

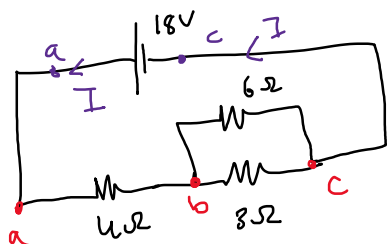


series





$$R_{eq} = \left(\frac{R_1 R_2}{R_1 + R_2} \right) + \left(\frac{R_3 R_4}{R_3 + R_4} \right) = R_{ac}$$



$R_{eq} = ?$ I on each resistor?
Power of the circuit?
Power of each resistor.

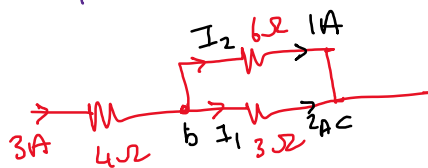
$V_{ac} = ?$ $V_{bc} = ?$

$\rightarrow 18V$

$$R_{eq} = 4\Omega + (3\Omega || 6\Omega) = 4\Omega + \frac{3(6)}{3+6} = 6\Omega$$

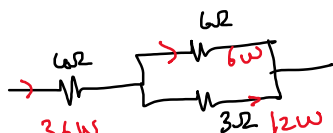
$$I = \frac{\Delta V}{R_{eq}} = \frac{18}{6} = 3A$$

$$\text{Power of circuit } I \Delta V = P = 18(3) = 54W \checkmark$$



$$V_{bc} = I_1 3 = I_2 6 = 6V \quad I_1 + I_2 = 3A$$

$\downarrow$ (2A) $\downarrow$ (1A)

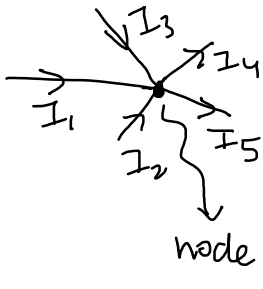


$$12V + 6V = 18V \checkmark$$

$$36 + 6 + 12 = 54W \checkmark$$

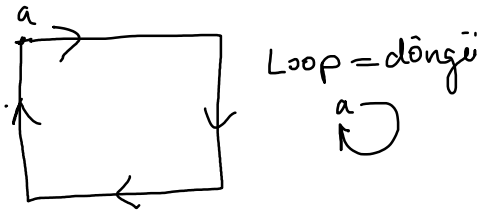
R	I	ΔV	P
4Ω	3A	12V	36W
6Ω	1A	6V	6W
3Ω	2A	6V	12W

KIRCHHOFF RULES



at a node; $\sum$ incoming current equal to total
total outgoing currents

$$\sum I_i = \sum I_o$$



* in a loop potential difference should be zero!

$$\Delta V = V_f - V_i = V_a - V_a = 0$$

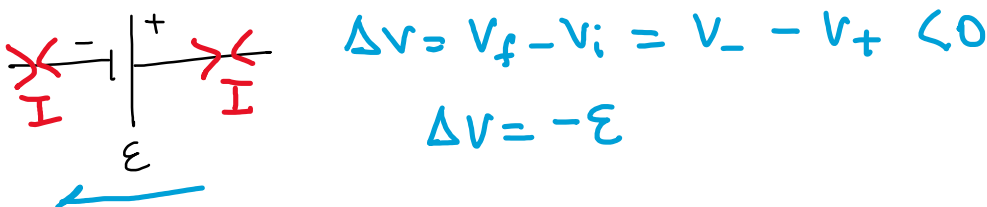
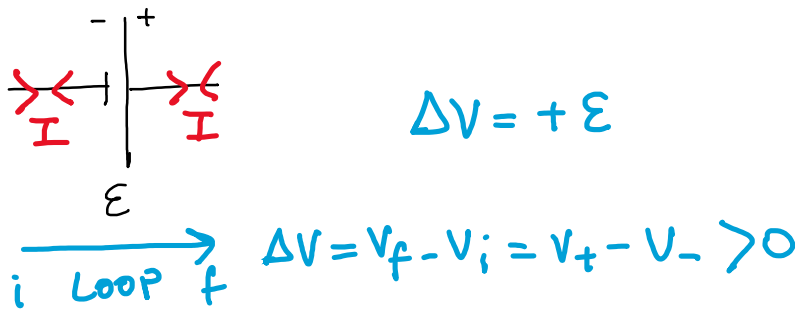
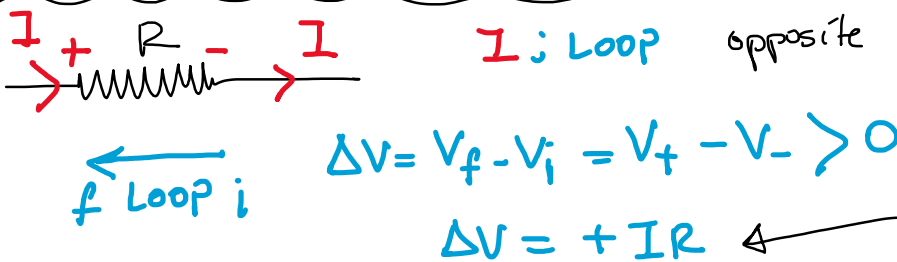
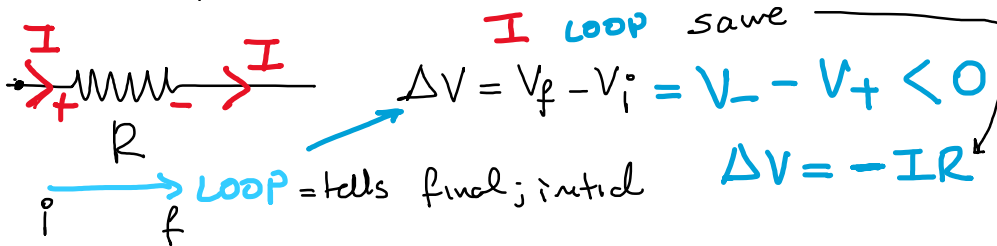
clockwise CW

$$\Delta V = V_f - V_i = V_a - V_a = 0$$

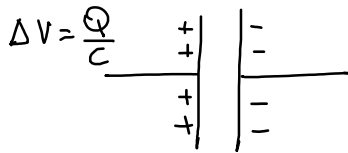
counter-clockwise CCW

DC = direct current

R Resistors
E Batteries
C Capacitors



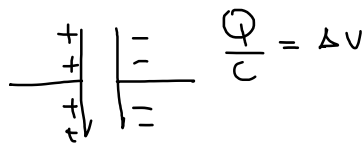
f-loop i



Loop

$$\Delta V = V_- - V_+ < 0$$

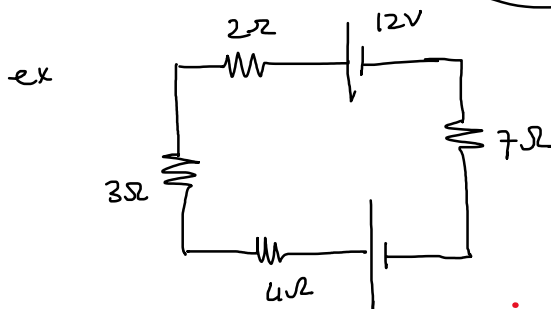
$$-\frac{Q}{C} = \Delta V$$



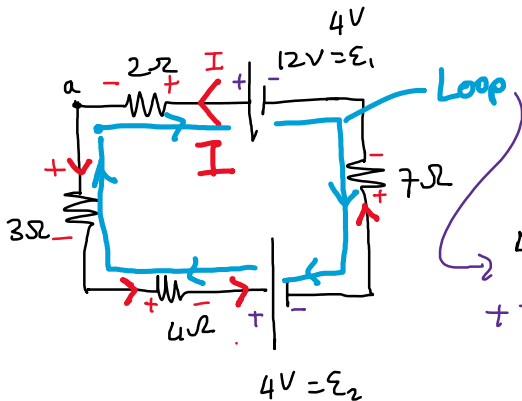
f-loop i

$$\Delta V = V_+ - V_- = V_+ - V_- > 0$$

$$\Delta V = +\frac{Q}{C}$$



$I = ?$ direction of current not given, we choose $\vec{i}$.



node = 0

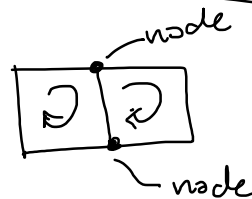
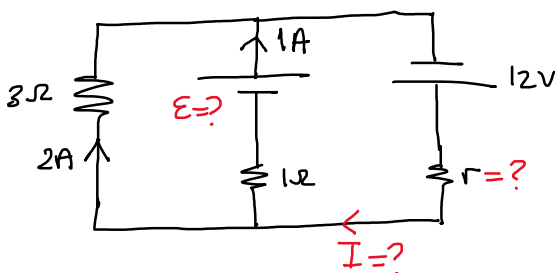
single Loop

$$\Delta V_{a \rightarrow a} = 0$$

$$\Delta V_{2\Omega} + \Delta V_{\epsilon_1} + \Delta V_{7\Omega} + \Delta V_{\epsilon_2} + \Delta V_{4\Omega} + \Delta V_{3\Omega} = 0$$

$$+I2 + (-12) + I7 + (4) + I4 + I3 =$$

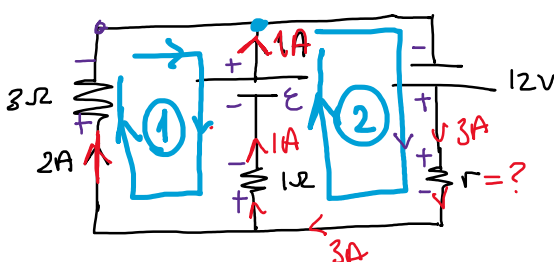
$$-8 + 16I = 0 \quad I = \frac{8}{16} = 0.5 \text{ A} \checkmark$$



$$\sum I_{in} = \sum I_{out}$$

$$I = 1 + 2$$

$$I = 3 \text{ A}$$

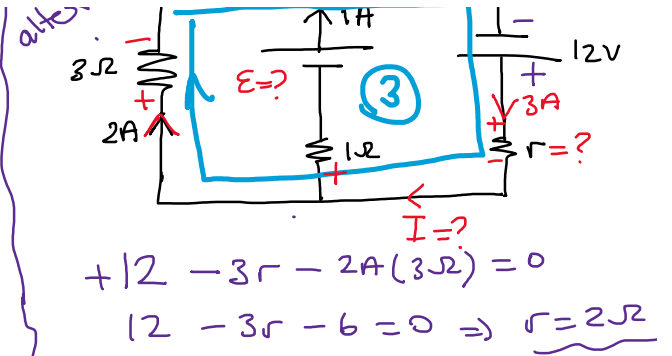


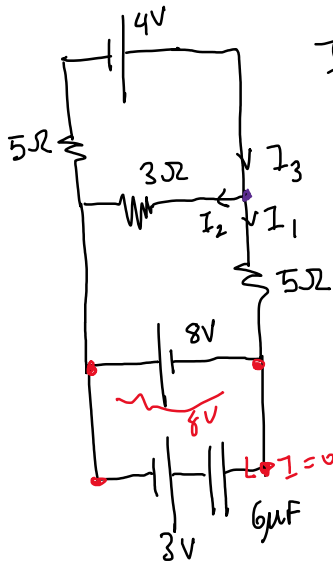
$$\textcircled{1} -\epsilon + (1\text{A})(1\Omega) - (2\text{A})(3\Omega) = 0$$

$$-\epsilon + 1 - 6 = 0$$

$$\epsilon = -5 \text{ V}$$

(2) $+12 - 3r - (1A)(1\Omega) + \epsilon = 0$
 $12 - 1 + (-5) = 3r$
 $\frac{6}{3} = r = 2\Omega$ ✓
 done ✓

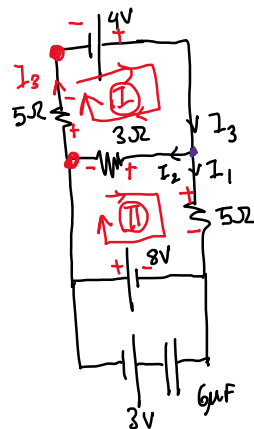
alter: 
 $+12 - 3r - 2A(3\Omega) = 0$
 $12 - 3r - 6 = 0 \Rightarrow r = 2\Omega$



$I_1, I_2, I_3 = ?$

$V_C = ?$ pot. of capacitor.

KCL $\Rightarrow$ node $I_3 = I_1 + I_2$ ①



①

$4 - 3I_2 - 5I_3 = 0$

$3I_2 + 5I_3 = 4$ ②

②

$+3I_2 - 5I_1 + 8 = 0$

$5I_1 - 3I_2 = 8$ ③

$3I_2 + 5I_1 + 5I_2 = 4$

$8I_2 + 5I_1 = 4$ ②

$-3I_2 + 5I_1 = 8$ ③

$[2-3] \Rightarrow 8I_2 + 3I_2 = -4$

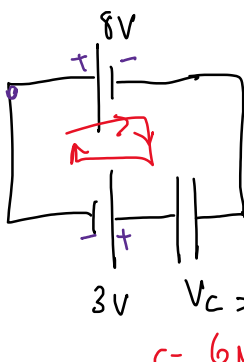
$I_2 = -\frac{4}{11} A$

$-3\left(-\frac{4}{11}\right) + 5I_1 = 8 \Rightarrow I_1 = \frac{1}{5}\left(8 - \frac{12}{11}\right)$

$I_1 = \frac{76}{55} A$

$I_3 = I_1 + I_2$

$= \frac{76}{55} + \frac{-20}{55} = \frac{56}{55} A$

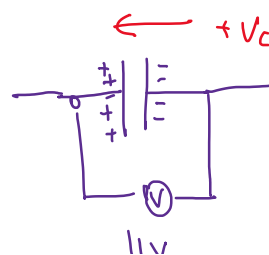


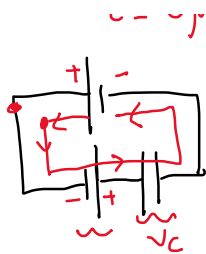
$-8 + V_C - 3 = 0$

$V_C = 11V$

$Q = ?$

$\frac{Q}{C} = V$





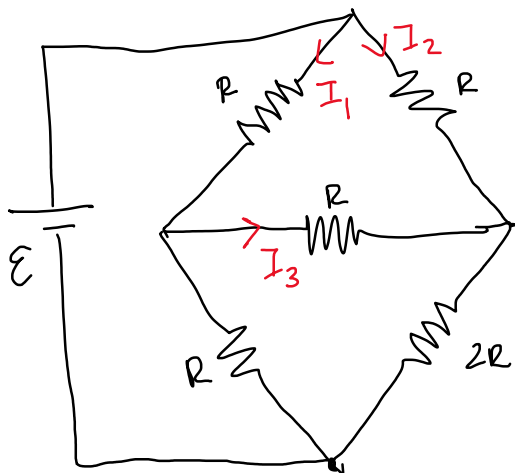
$$Q = CV = 6\mu F (11V) = \underline{\underline{66\mu C}}$$

$$+3 + V_c + 8 = 0$$

$$V_c = \underline{\underline{-11V}}$$

$$\begin{array}{|c|c|} \hline + & - \\ \hline \end{array} -11V$$

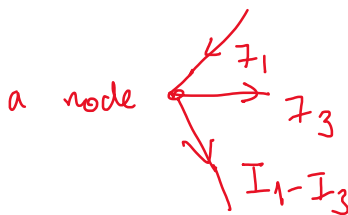
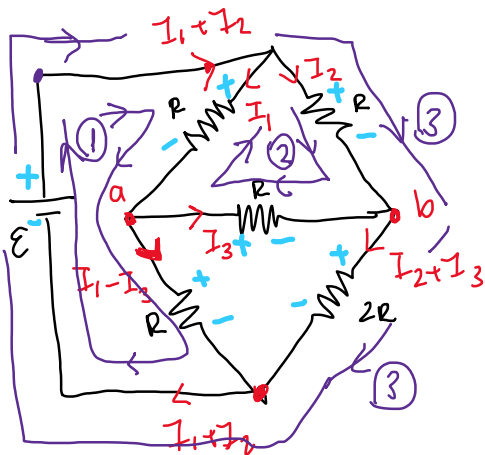
$$\begin{array}{|c|c|} \hline - & + \\ \hline \end{array} +11V \checkmark$$



$$R = 1\Omega \quad E = 13V$$

$$I_1, I_2, I_3 = ?$$

R_{eq} = equivalent resistance of circuit



①

$$-I_1 + (- (I_1 - I_3)) + E = 0$$

$$-2I_1 + I_3 + E = 0 \quad \text{①}$$

$$\text{② } -I_2 + I_3 + I_1 = 0$$

$$\text{③ } -I_2 + [-2(I_2 + I_3)] + E = 0$$

$$-3I_2 - 2I_3 + E = 0$$

$$\text{① } 2I_1 - I_3 = E = 13$$

$$\text{② } I_1 - I_2 + I_3 = 0$$

$$\text{③ } 3I_2 + 2I_3 = E = 13$$

$$I_2 = I_1 + I_3$$

$$3I_1 + 3I_3 + 2I_3 = 13$$

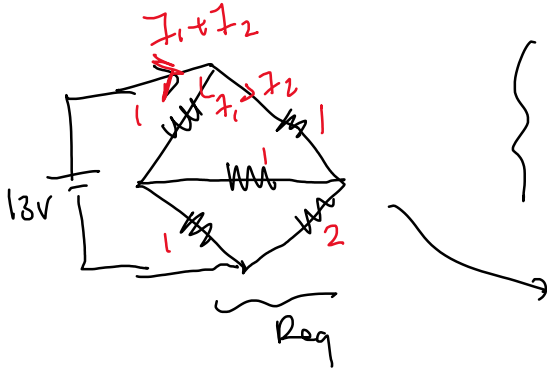
$$3I_1 + 5I_3 = 13$$

$$\begin{array}{l} \times 5 (2I_1 - I_3 = 13) \\ + \quad 3I_1 + 5I_3 = 13 \end{array}$$

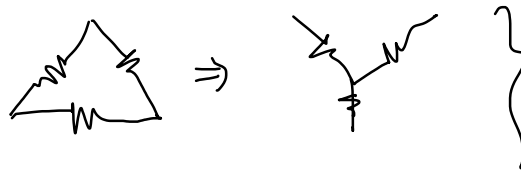
$$13I_1 = 6(13) \quad \boxed{I_1 = 6A}$$

$$2(6) - I_3 = 13 \quad \boxed{I_3 = -1A}$$

$$I_2 = I_1 + I_3 = 6 + (-1) = 5A$$

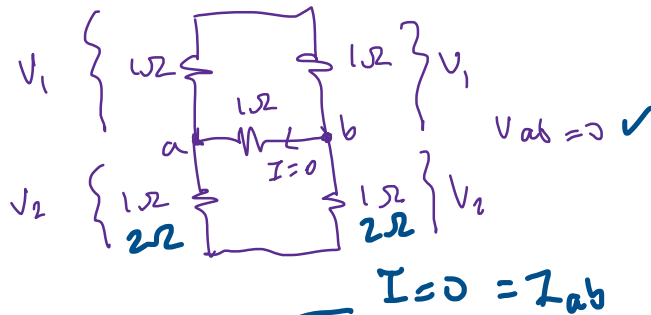
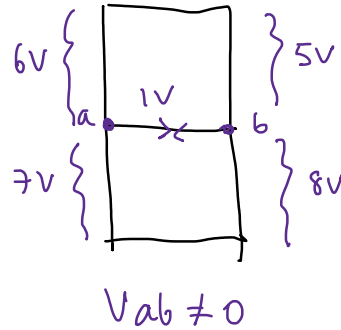
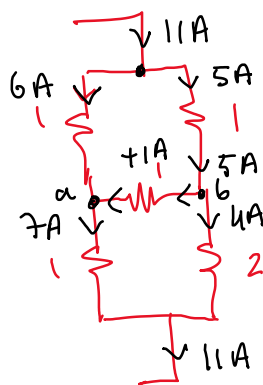
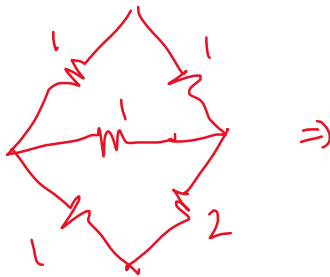


$$R_{eq} = \frac{13}{11} \Omega$$



$$\mathcal{E} = R_{eq} (I_1 + I_2) \quad (R_{eq})(I_1 + I_2) = \mathcal{E}$$

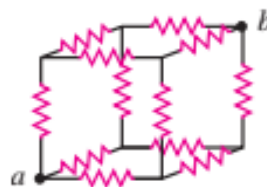
$$R_{eq} (6 + 5) = 13$$



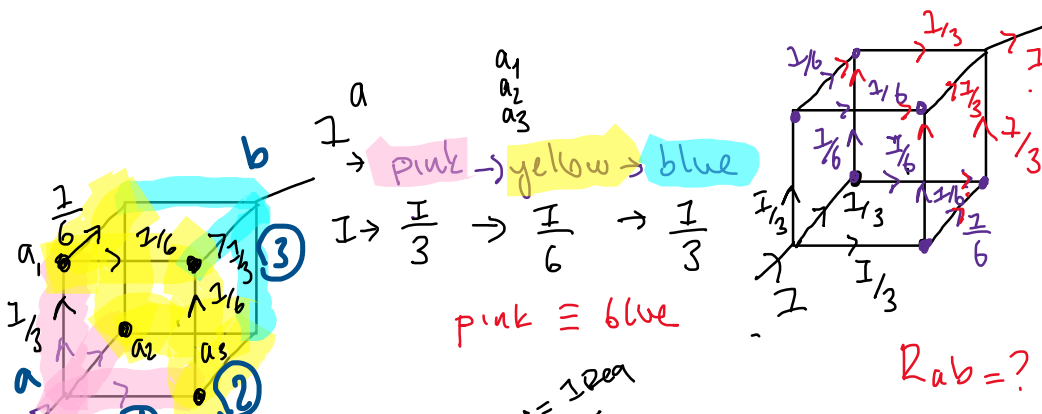
$$I = 0 = I_{ab}$$

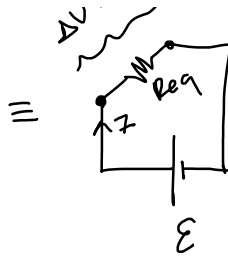
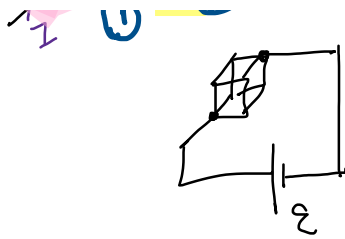
26.92 ... Suppose a resistor R lies along each edge of a cube (12 resistors in all) with connections at the corners. Find the equivalent resistance between two diagonally opposite corners of the cube (points a and b in Fig. P26.92).

Figure P26.92

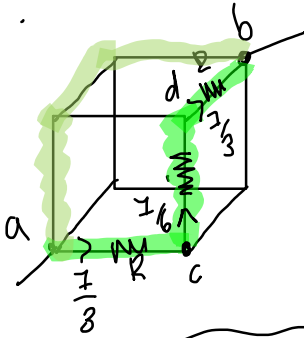


$$R_{ab} = ?$$





$$V_{ab} = \Delta V = I R_{eq}$$

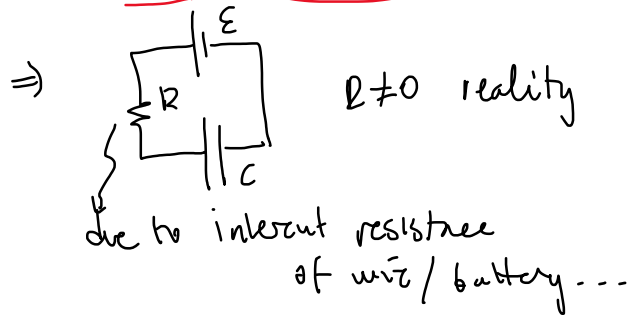


$$V_{ab} = V_{ac} + V_{cd} + V_{db}$$

$$V_{ab} = \frac{I}{3} R + \frac{I}{6} R + \frac{I}{3} R = \frac{5IR}{6} = I R_{eq}$$

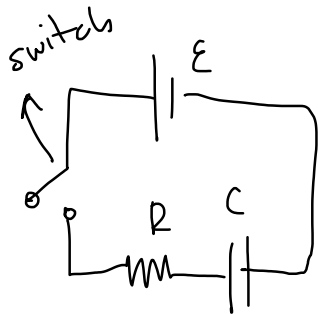
$$R_{eq} = \frac{5}{6} R$$

DC circuits



$R \neq 0$ reality

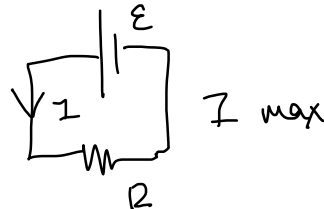
RC circuits have "time constants" $\Rightarrow$ traffic lights
blinking lights
windshield wipers (sleeeek)



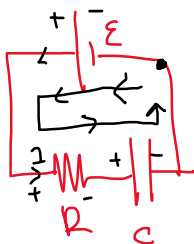
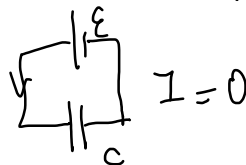
switch on $\Rightarrow$ C starts charging up.

wait long enough C is full
there is no current on the circuit.

$$t \approx 0s = t = 0s^+$$



?? $0 \leq t \ll \log$
 $t \gg 0$



$$\Delta V = 0$$

$$+\mathcal{E} - V_R - V_C = 0$$

$$\mathcal{E} - IR - \frac{q}{C} = 0$$

$$I = \frac{dq}{dt}$$

$$\left| \mathcal{E} - \frac{dq}{dt} R - \frac{q}{C} = 0 \right| \text{ diff. equ.}$$

$$\cancel{R} \frac{dq}{dt} = \left(\mathcal{E} - \frac{q}{C} \right) \frac{1}{\cancel{R}} \Rightarrow \frac{dq}{\left(\mathcal{E} - \frac{q}{C} \right) \frac{1}{R}} = dt$$

$$\left(\mathcal{E} - \frac{q}{C} \right) \frac{1}{R} = \frac{\mathcal{E}}{R} - \frac{q}{RC} = \frac{\mathcal{E}C - q}{RC}$$

$$\frac{dq}{q - \mathcal{E}C} = - \frac{dt}{RC}$$

$$\boxed{\frac{dq}{(\mathcal{E}C - q)} = \frac{dt}{RC}}$$

$RC = \{\text{sec}\} ?!$

$$q - \mathcal{E}C = u \quad \begin{cases} dq = du \\ \int \frac{du}{u} = \int -\frac{dt}{RC} \end{cases}$$

$$\ln u = \ln(q - \mathcal{E}C) \quad \left| \begin{array}{l} q_f = q(t) \\ q_i = 0 \end{array} \right| = - \frac{\Delta t}{RC} \quad \left| \begin{array}{l} t_f = t \\ t_i = 0 \end{array} \right|$$

$$t_i = 0s; \quad @ \quad 0s \quad q_i = 0 \quad \leadsto \quad q_f = q(t) \quad t_f = t$$

$$\ln(q(t) - \mathcal{E}C) - \ln(0 - \mathcal{E}C) = - \frac{(t - 0)}{RC}$$

$$\ln \left(\frac{q(t) - \mathcal{E}C}{-\mathcal{E}C} \right) = - \frac{t}{RC}$$

$$\ln \left(\frac{\mathcal{E}C - q(t)}{\mathcal{E}C} \right) = - \frac{t}{RC} \Rightarrow \frac{\mathcal{E}C - q(t)}{\mathcal{E}C} = e^{-t/RC}$$

$$\mathcal{E}C - q(t) = \mathcal{E}C e^{-t/RC}$$

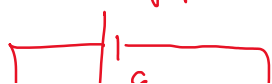
$$\mathcal{E}C - \mathcal{E}C e^{-t/RC} = q(t) = \mathcal{E}C (1 - e^{-t/RC})$$

$$\hookrightarrow I = \frac{dq}{dt} = \mathcal{E}C \left[0 - \left(-\frac{1}{RC} \right) e^{-t/RC} \right]$$

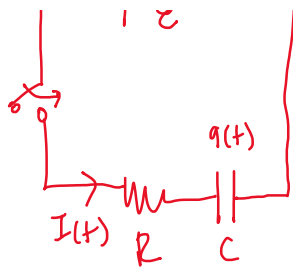
$$I(t) = \frac{\mathcal{E}}{R} e^{-t/RC}$$

$$\leftarrow I(t) = \frac{\mathcal{E}C}{RC} e^{-t/RC}$$

charging of capacitor in a RC circuit



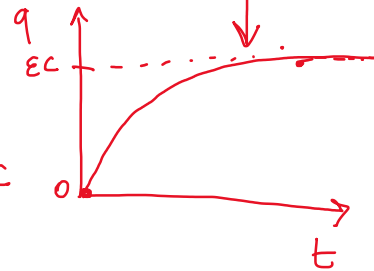
$$q(t) = \mathcal{E}C \left(1 - e^{-\frac{t}{RC}} \right)$$



$$I(t) = \frac{\varepsilon}{R} e^{-t/RC}$$

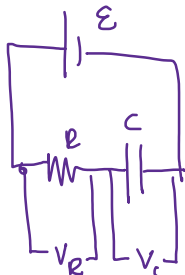
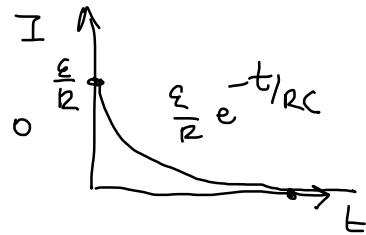
$$t=0 \quad q(0) = \varepsilon C (1 - e^{-0/RC}) = 0$$

$$t \gg 0 \quad q(t \gg 0) = \varepsilon C (1 - e^{-\infty/RC}) = \varepsilon C$$



$$I(t) \quad t=0 \quad I(0) = \frac{\varepsilon}{R} e^{-0/RC} = \frac{\varepsilon}{R}$$

$$t \gg 0 \quad I(t \gg 0) = \frac{\varepsilon}{R} e^{-\infty/RC} = 0$$

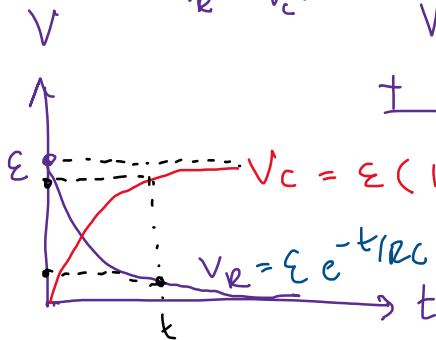


$$\varepsilon - V_R - V_C = 0$$

$$\varepsilon = V_R + V_C$$

$$V_R(t) = I(t) R = \frac{\varepsilon}{R} e^{-t/RC} R = \varepsilon e^{-t/RC}$$

$$V_C(t) = \frac{q(t)}{C} = \frac{\varepsilon C}{C} (1 - e^{-t/RC}) = \varepsilon (1 - e^{-t/RC})$$



ex) $\varepsilon = 12V$ $R = 100\Omega$ $C = 20\mu F$

C is charging up.

a) what is the time constant? $\tau = \tau_{RC} = \tau_{RC}$

$$\tau = RC = 100\Omega \cdot 20\mu F = 2000\mu s$$

b) charge on C @ $t = 1ms$?

$$q(t=1ms) = \underbrace{\varepsilon}_{12V} \underbrace{C}_{20\mu F} (1 - e^{-t/RC}) = 240\mu C (1 - e^{-t/2ms})$$

$$240\mu C (1 - e^{-1/2})$$

$$c) I(t=1ms) = \frac{\varepsilon}{R} e^{-t/RC}$$

$$= 94.4\mu C$$

$$= \frac{12}{100} A e^{-1/2} = 0.07 A = 70mA$$

d) $V_R(t=1\text{ms}) = I(t)R = 70\text{mA}(100\Omega) = 7\text{V}$

e) at what time is the capacitor half-full?
yarı dolmuş zamanı?

$$q(t) = \underbrace{\epsilon C}_{Q_{\max}} (1 - e^{-t/RC}) \Rightarrow \frac{Q_{\max}}{2} = Q_{\max} (1 - e^{-t/2\text{ms}})$$

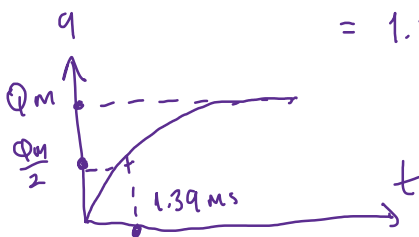
$$\frac{-t}{2\text{ms}} = -0.69$$

$$t = 0.69(2\text{ms}) = 1.39\text{ms}$$

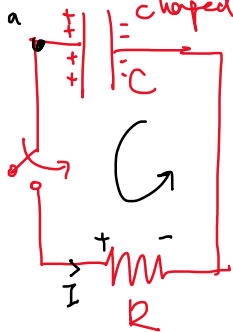
$$\frac{1}{2} = 1 - e^{-t/2\text{ms}}$$

$$e^{-t/2\text{ms}} = \frac{1}{2}$$

$$-\frac{t}{2\text{ms}} = \underbrace{\ln\left(\frac{1}{2}\right)}_{<0} = -0.69$$



Discharging of a charged capacitor in RC circuit



$$\Delta V_{a \rightarrow b} = V_C - V_R = 0$$

$$\frac{q}{C} - IR = 0$$

$$I = \frac{dq}{dt}$$

$$\frac{q}{C} - \frac{dq}{dt} R = 0$$

$$\frac{q}{RC} = \frac{dq}{dt}$$

$$\frac{dq}{q} = \frac{dt}{RC}$$

$$dq = q_f - q_i < 0$$

$$\frac{-|dq|}{q} = \frac{dt}{RC} \Rightarrow \int_{q_i}^{q_f} \frac{dq}{q} = \int_0^t -\frac{dt}{RC}$$

$$\ln(q) \Big|_{q_i}^{q_f} = -\frac{\Delta t}{RC} \Big|_0^t$$

$$\ln q_f - \ln q_i = -\frac{(t-0)}{RC} = -\frac{t}{RC}$$

$$\ln\left(\frac{q(t)}{q_0}\right) = -\frac{t}{RC} \Rightarrow q(t) = q_0 e^{-t/RC}$$

$$I(t) = \frac{dq}{dt} = q_0 \left(-\frac{1}{RC}\right) e^{-t/RC}$$

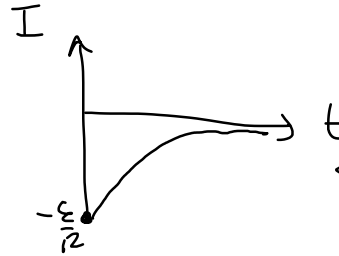
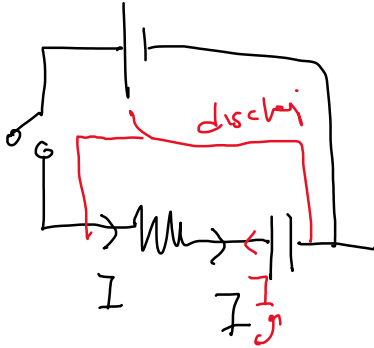
$$q_0 = q_{\max} = \epsilon C$$

$$I(t) = -\frac{q_0}{RC} e^{-t/RC}$$

$$q_0 = \varepsilon C$$

$$I(t) = -\frac{\varepsilon}{R} e^{-t/RC}$$

$\Rightarrow$ opposite direction of charging



charging - discharge cycles

